



EXECUTIVE INSIGHTS

Digital Pathology Demand and Adoption Trends — Insights From L.E.K.'s US Clinical Diagnostic Lab Survey

Introduction

Anatomic pathology in the U.S. is gradually shifting toward digitally enabled workflows as whole-slide imaging (WSI) expands and artificial intelligence (AI)-enabled image analysis tools gain traction. For suppliers, this transition raises a practical set of commercial questions: How quickly will different customer segments scale, which barriers and adoption drivers matter most and which use cases will deliver the clearest value to customers?

To inform key market gaps and commercial opportunities, L.E.K. Consulting's U.S. Diagnostic Lab Survey captures perspectives from 100+ hospital and multi-specialty reference lab leaders on key trends in digital pathology adoption and use cases.

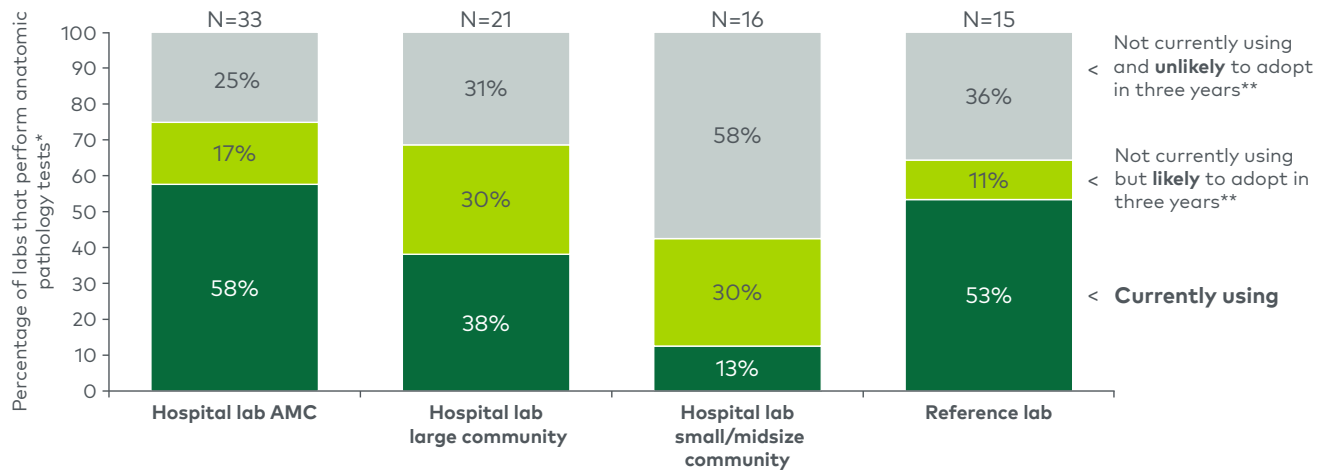
Key trends

Community hospitals will drive the next wave of digital pathology adoption

Digital pathology adoption is now concentrated in academic medical centers (AMCs) and reference labs, where approximately 60% of AMCs and 50% of surveyed reference labs report current WSI use for clinical purposes (see Figure 1).

Figure 1

Current and expected near-term digital pathology adoption



*Survey questions: Which, if any, of the following emerging technology platforms does your lab (single lab site) currently use in-house — d. Pathology whole-slide imaging (e.g., Leica Biosystems Aperio GT450)? Of all the emerging technology platforms that you are NOT using today, how likely are you to purchase one in the next three years?

**Likelihood of adoption has been over-statement adjusted using the following method: 80% of respondents choosing "definitely will" will adopt, 60% of respondents choosing "probably will" will adopt, 25% of respondents choosing "might or might not" will adopt, 0% of respondents choosing "probably will NOT" or "definitely will NOT" will adopt

Note: AMC=academic medical center

Source: L.E.K. 2025 Diagnostic Lab Survey

Community hospitals are positioned to lead the next wave of adoption; penetration could nearly double in large community hospitals and triple in small/midsize community hospitals over the next three years if labs act on their stated intentions. This reflects a growing recognition of digital pathology's value and its expansion into broader care settings.

Cost and integration remain the biggest barriers to broader uptake

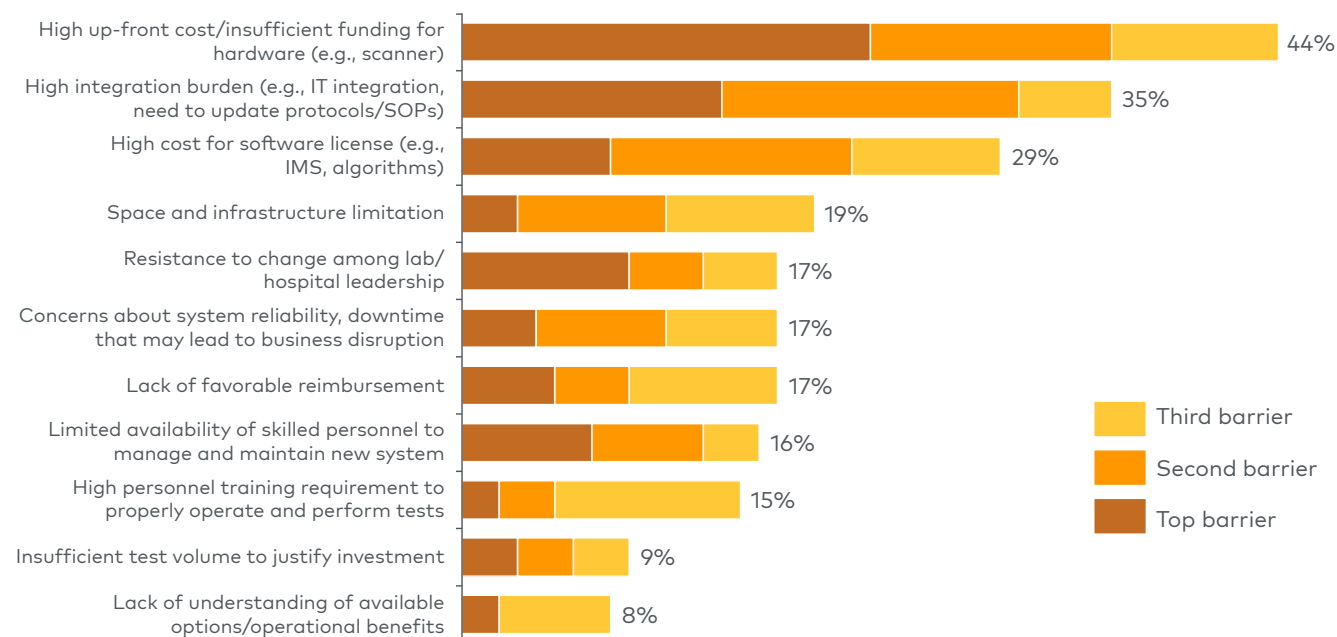
The most frequently cited barriers to broader digital pathology adoption are cost — primarily hardware and to a lesser extent software — and integration burden (see Figure 2), suggesting that for many labs the return-on-investment (ROI) case is not yet clearly established.

Figure 2

Key perceived barriers to digital pathology adoption

Top three barriers to adopting digital pathology more broadly

Percentage of selected labs routinely offering pathology, N=84*



*Survey question: What, if any from list below, are the key barriers to adopting digital pathology more broadly in your lab (single lab site)?

Note: IT=information technology; SOPs=standard operating procedures; IMS=image management system

Source: L.E.K. 2025 Diagnostic Lab Survey

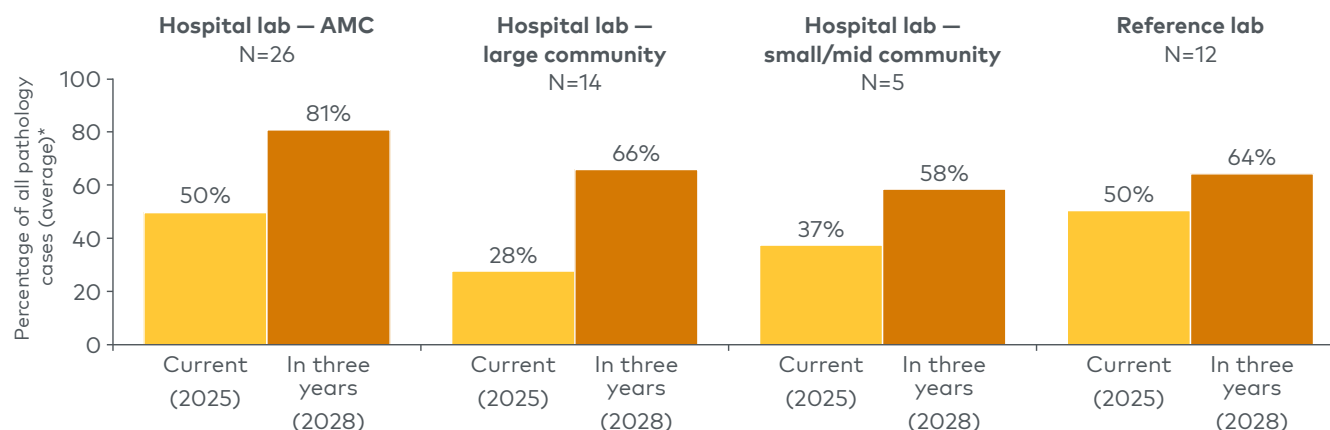
Beyond technical performance and reliability, winning solutions for new adopters will need to minimize implementation friction (e.g., information technology (IT) integration, standard operating procedure updates, personnel training) and make a strong case for ROI, not only through flexible commercial structures but also by demonstrating measurable gains in productivity.

Digitization is expanding among current users but stops short of 100%

Among labs currently using digital pathology, average utilization levels remain well below full digitization across all cases. AMC and reference lab users digitize approximately 50% of cases on average today, while large community hospital users digitize less than 30% (see Figure 3), reflecting selective use rather than end-to-end workflow redesign.

Figure 3

Digital pathology utilization among current adopters, current and in 3 years (2025-28), by lab type



*Survey questions: What percentage of all pathology cases performed in-house in your lab (single lab site) are currently digitized (i.e., slides scanned and digitized)? What do you expect this percentage to be in three years?

Note: AMC=academic medical center

Source: L.E.K. 2025 Diagnostic Lab Survey

Across lab settings, current users expect to grow their digital pathology utilization materially over the next three years, with the largest increase anticipated in large community hospital labs. However, even AMCs expect to stop short of full digitization, at around 80%, suggesting that digitization may remain operationally impractical or of low incremental value for a subset of cases.

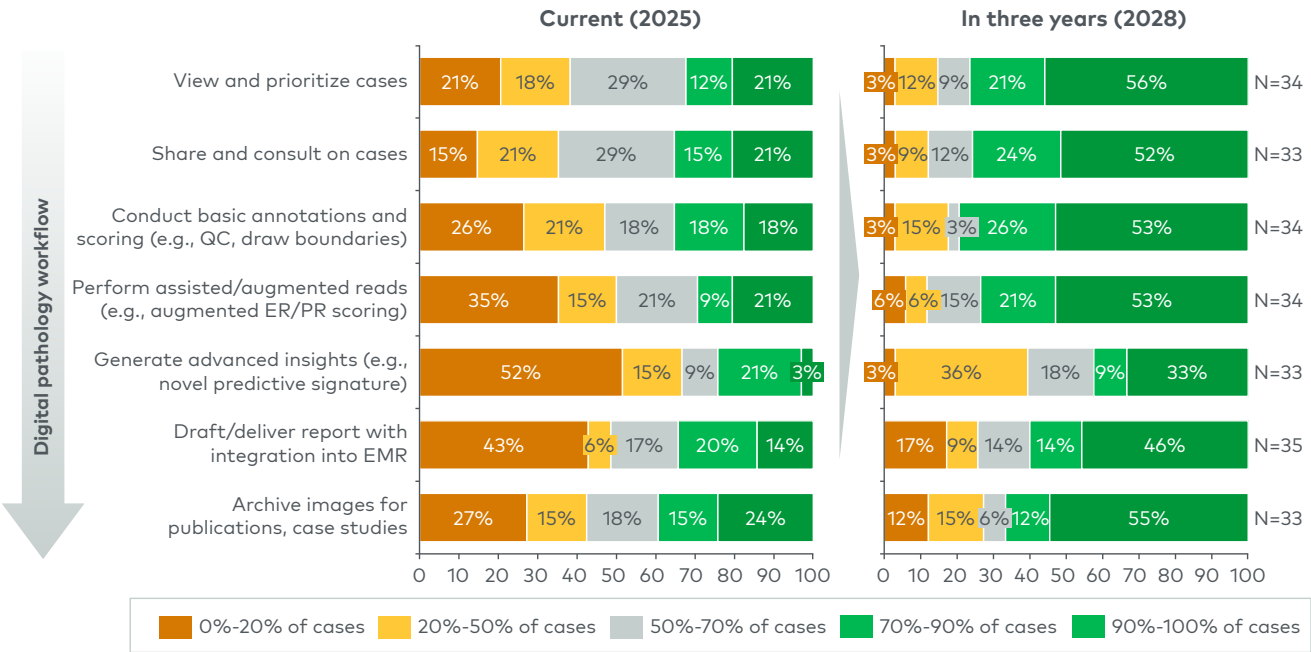
Use is growing across all jobs-to-be-done, most notably in assisted reads and advanced insights

Among current adopters, digital pathology is evolving from basic functions into a platform that orchestrates workflows and enables higher-value interpretation. Current use is mainly anchored in operational use cases (case viewing/prioritization, sharing/consults, archiving for teaching/research) and routine annotations and scoring (see Figure 4). These use cases are expected to remain the highest penetrated applications for the near future, with over half of experts anticipating utilization in more than 90% of digitized cases in three years. While report drafting/delivery lags other operational use cases today, it is expected to increase substantially as digital pathology embeds further into reporting workflows. Although AI-generated reports remain at an early stage, researchers have demonstrated pathology report generation from WSI (e.g., HistoGPT), and there is an emerging regulatory precedent in radiology (July 2026 FDA clearance of DeepHealth Breast Ultrasound).

Figure 4

Digital pathology use cases among current adopters, current and in 3 years (2025-28)

Pathology jobs-to-be-done performed using image viewer or IMS*
Percentage of digitized pathology cases**



*Survey questions: Of all digitized pathology cases (i.e., cases that are already scanned and digitized), what percentage of cases are performing each of the following tasks using an image viewer or IMS? What do you expect the percentage to be in three years?

**Respondents who answered "I don't know" were excluded from analysis

Note: QC=quality control; ER/PR=estrogen receptor/progesterone receptor; EMR=electronic medical record; IMS=image management system

Source: L.E.K. 2025 Diagnostic Lab Survey

Higher-value uses, such as assisted/augmented reads, are also expected to grow significantly, reflecting greater willingness to use algorithm-enabled quantification when it improves consistency, reduces hands-on effort or sustains throughput amid staffing constraints. The prospect of companion diagnostics (CDx) may be a further driver; although no computational pathology CDx has been approved to date, some labs may already be anticipating one as the pipeline for targeted therapies, particularly ADCs, continues to grow. Algorithms can potentially improve therapy response prediction, including by enabling continuous scoring (e.g., TROP2 QCS) in place of today's semiquantitative approach, accurately detecting ultralow expression of targets (e.g., HER2), quantifying surface vs intracellular expression, or measuring a radiopharmaceutical's effective range (e.g., for alpha particle therapy). Advanced insights remain the most nascent use case, but users anticipate a meaningful step-change in use, reflecting demand for capabilities beyond human visual interpretation and optimism that algorithms can deliver clinical value.

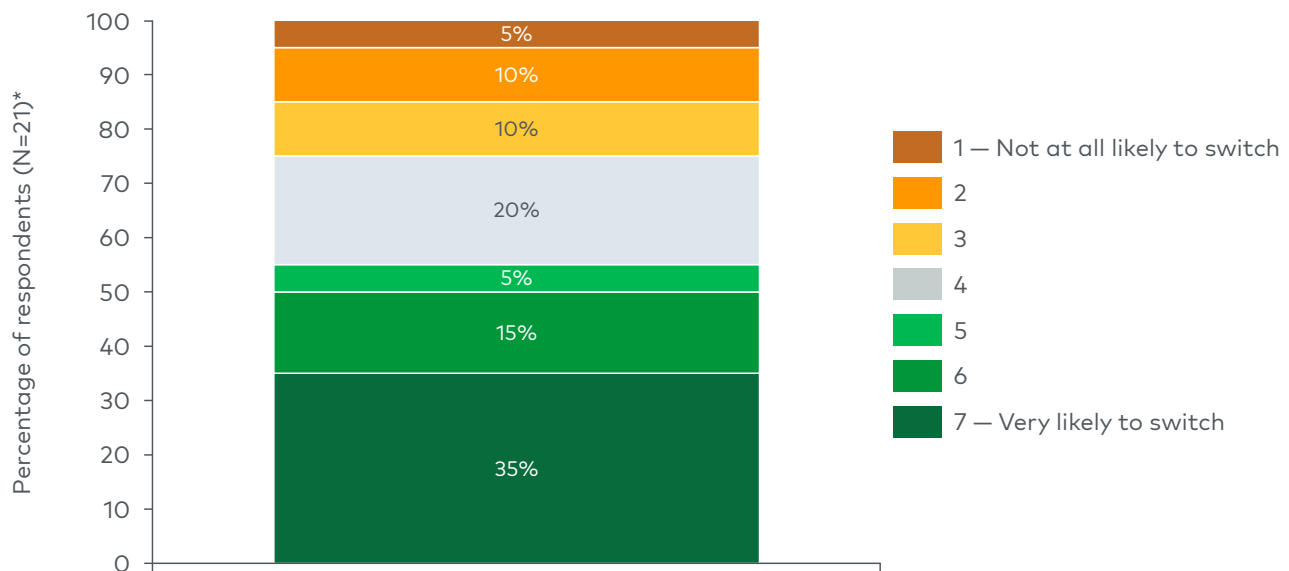
Current users expect near-term slide scanner purchases, with over half open to switching vendors

More than 80% of current WSI users expect to purchase a new instrument within five years, with more than 30% of them anticipating making the purchase within the next year. Most planned purchases are driven by capacity expansion rather than pure replacement or consolidation, signaling underlying growth in demand among current adopters.

Figure 5

Vendor switching intent among users planning to purchase a new slide scanner

Possibility of **switching instrument vendor****



*Survey question: For each instrument type you are planning to purchase in the next five years, how likely are you to switch instrument vendors for your next purchase? — Pathology slide scanner

**Respondents who answered "I don't know" were excluded from analysis; all selected respondents are planning to purchase new pathology scanners within five years

Source: L.E.K. 2025 Diagnostic Lab Survey

More than half of current users planning a new scanner purchase are open to switching vendors. That group includes 35% who are "very likely" to switch — higher than any other major lab instrument type included in our survey (see Figure 3 and our previous EI¹ for a cross-modality comparison). This may reflect a gap between labs' initial platform choices and their evolving needs as operations grow (e.g., throughput, uptime, workflow integration, service responsiveness). For suppliers, users' willingness to switch presents a share capture opportunity, which will favor those that can best address segment-specific pain points and capacity needs.

Implications for OEMs and lab suppliers

As adoption broadens to the next wave of U.S. labs, the market will increasingly favor suppliers that pair a credible ROI case with low-friction implementation and support for labs across applications, from operational use cases toward assisted reads and advanced insights.

Build an ROI case with customers

Despite strong interest in digital pathology, hardware and software costs remain two out of the top three adoption barriers. Suppliers can work with customers to quantify value, drawing on labs' operational data to assess total cost of ownership and potential operational gains in terms that matter to them (e.g., reduced slide transport costs, faster consults, less time spent on slide-heavy cases). Flexible commercial structures (e.g., subscription or consumption-based models, trial periods that let labs prove workflow benefit before scaling) can then make the investment less risky for labs.

Reduce integration and validation burden

Integration burden, another top barrier, can be especially challenging in hospital settings with custom laboratory information system/electronic medical record configurations and limited IT bandwidth. Solutions that get labs up and running faster through prebuilt integrations and hands-on support can directly address this pain point. Easing the validation burden can also help: Vendor-provided templates, protocols and training, aligned with College of American Pathologists guidance, can reduce the internal effort required of labs.

Capture share now while openness to switching is high

High switching intent among current users reflects a still-fluid market where some labs may not yet be deeply integrated and both lab needs and vendor offerings are still evolving. As integration deepens, switching becomes more costly for customers; therefore, the window for suppliers to win accounts and build lasting positions is now. Vendors can invest in facilitating low-risk migration (e.g., playbooks to minimize disruption and data and configuration conversion tools) to win labs over, then embed more deeply into lab workflows to protect those accounts over time.

Serve operational needs today while building for the higher-value future

Near-term demand centers on operational use cases. Suppliers should lead with measurable efficiency wins (e.g., time saved per case, turnaround time improvement). As the market matures, demand for assisted reads and advanced insights will grow, and suppliers will need

to deliver those tools to avoid displacement. Algorithm vendors, for their part, will have to differentiate on demonstrated utility in improving lab efficiency or clinical outcomes, depending on the application, and on how easily they embed into workflows.

To discuss these findings and translate them into commercial actions across products, services and informatics, please **reach out** to our Diagnostics team.

A special thank-you to the Healthcare Insights Center for its contributions to this edition of *Executive Insights*.

Note: AI was used to support the drafting of this piece.

Endnote

¹LEK.com, "Clinical Diagnostic Testing Trends Across Key Modalities: Insights From L.E.K.'s US Diagnostic Lab Survey (2025)." <https://www.lek.com/insights/life-sciences-pharma/clinical-diagnostic-testing-trends-across-key-modalities-insights>

About the Authors



Catia Verbeke

Catia Verbeke is a Managing Director in L.E.K. Consulting's San Francisco office and a member of the firm's Life Sciences practice. Catia specializes in precision medicine, diagnostics and advanced research tools. She has advised on growth strategy, commercial planning and business development, with a focus on helping biopharma, diagnostics, and research tools clients navigate complex markets and innovate strategies.



Tian Han

Tian Han is a Managing Director and Partner in L.E.K. Consulting's Los Angeles office and a member of the firm's Life Sciences practice. Tian specializes in precision medicine, advanced diagnostics, life science research tools and advanced therapeutic bioprocessing. He advises clients on corporate and business strategy, product strategy, commercial planning and transaction support.



Alex Vadas

Alex Vadas, Ph.D. is a Managing Director and Partner in L.E.K. Consulting's Healthcare practice and co-leads the Life Sciences Enablers practice. Alex has worked with many financial and strategic clients from venture-backed to global multinationals in corporate strategy, product strategy and planning, as well as transaction support. He also specializes in life sciences tools and technologies, diagnostics and precision medicine, and advanced therapy bioprocessing and manufacturing.

**Yilun Deng**

Yilun Deng is a Consultant in L.E.K. Consulting's Houston office, dedicated to the Life Sciences sector. Yilun's work centers on precision medicine, diagnostics, research tools and other critical enablers of successful therapeutic innovation. She earned a PhD in cancer biology from UT Health San Antonio and has extensive academic experience in oncology and cancer immunology.

**Jenny Mackey**

Jenny Mackey is the Director of L.E.K. Consulting's Healthcare Insights Center. Prior to this, Jenny was a Principal in L.E.K.'s Biopharma practice, where she advised clients on a range of issues including R&D portfolio prioritization, new product planning, forecasting and valuation, and organizational performance and development.

About L.E.K. Consulting

L.E.K. Consulting is a premier global strategy consulting firm that partners with clients to solve their most critical business challenges and deliver high-impact outcomes. We bring deep industry expertise and rigorous, evidence-based insight to global corporations, growth companies and private equity investors, helping them turn strategy into action. Founded more than 40 years ago, L.E.K. now operates across 27 offices worldwide. Learn more at lek.com.

L.E.K. Consulting is a registered trademark of L.E.K. Consulting LLC. All other products and brands mentioned in this document are properties of their respective owners. © 2026 L.E.K. Consulting LLC